Cardio Digital Twin: system architecture

Two streams, one daily status per patient. Blue = real data; amber = simulated data; teal = where the streams are fused.

Stream 1: static clinical data (REAL)

PTB-XL v1.0.3 (PhysioNet, CC BY 4.0)
21,799 ten-second 12-lead ECGs, 18,869 patients
Cardiologist labels, age, sex. Anonymised and open.

↓

Preparation (prepare_data.py)
Label: normal vs any of MI, ST/T, conduction, hypertrophy
Adults only; official patient-wise folds 1–8 / 9 / 10
Train 16,973 · validation 2,137 · test 2,145 · 500 → 100 Hz

↓

ECG model (PyTorch 1-D CNN)
4 conv blocks → P(abnormal ECG); 189,889 parameters
4 cross-fitted CNNs, probabilities averaged = risk score
Age and sex tested: no measurable gain, so the score is ECG-only

↓

Risk tier
High risk if score ≥ 0.511 (90% specificity on validation)
Test: AUROC 0.938 (95% CI 0.928–0.947); sensitivity 0.80, specificity 0.91

Fusion 2: twin state (twin_state.py)

One status per patient per day

	no alarm	alarm
Low risk	**Stable**	**Watch**
High risk	**Watch**	**Escalate**

Settings locked on validation before the test run

Fusion 1: tier sets alarm sensitivity

low risk: 0.70, k=4 · high risk: 0.55, k=3

Stream 2: continuous wearable data (SIMULATED)

Sources for the recipe
PMData (16 people, months of Fitbit data), MMASH (HRV)
Quer 2020 (resting HR) · Althoff 2017 (India steps) · Hannon 2025 (HRV)

↓

Simulator (fit_recipe.py, simulate_wearables.py)
Daily value = personal level + weekday + AR(1) wobble
60 days: resting HR, HRV, sleep, steps. Reads age, sex only
Episodes in 50%: resting HR +3/+5/+8/+12 bpm over 3–7 days

↓

Personal-baseline monitor (monitor.py)
z = direction × (value − median) / (1.4826 × MAD)
Baseline: the 28 days ending 8 days before today
Mean of 4 z, 3-day smoothing; alarm if above threshold k days running

↓

Daily alarm flag
Tuned on validation: ≤ 0.5 false alarms per patient-month
Real PMData check: 0.63 alarms per person-month (upper bound)

Doctor dashboard (Streamlit, dashboard.py)

Worklist
Escalate first; "alert for N days"; what changed in bpm, h, %; ECG risk; reviewed

Patient page
One-sentence summary; ECG card; 4 reading tiles; trend vs usual range

12-lead ECG
Real trace with optional saliency shading (sanity check: correlation 0.11)

Review and About
Note + mark as reviewed; how it decides, test results, limits; no treatment advice

Evaluation and integrity

- Rule written before every comparison (DECISIONS.md)
- All tuning on validation; test opened once (lock file)
- 9 pytest checks: no patient overlap, no look-ahead, locked settings, numbers recomputable
- Real-data false-alarm check; simple-rule baseline
- Fixed seeds; reproduced on a second machine